BIM472 IMAGE PROCESSING
HOMEWORK #1

I. Using Matlab (40 pts)

The goal of this problem set is to become familiar with basic Matlab commands, practice manipulating vectors and matrices, and try out basic image display and plotting functions. If you are unsure what a Matlab function does, check the reference manual (at the command line, type "help" and then the command name).

1. (Each 2 pts) Describe (in words where appropriate) the result of each of the following Matlab commands. Use the help command as needed but try to determine the output without entering the commands into Matlab. Do **not** submit a screenshot of the result of typing these commands.

   a. >> x = randperm(50);
   b. >> a = [1:2:10; 11:2:20]';
      >> b = a(2,:);
   c. >> f = randn(10,2);
      >> g = f(find(f < 1));
   d. >> x = 0.5.* ones(1,10);
      >> y = 0.5+zeros(1,length(x));
      >> z = x + y;
   e. >> a = [1:100];
      >> b = a([end:-1:1]);

2. (Each 5 pts) Suppose you are given a 100 x 100 matrix A representing a grayscale image. Write a few lines of code to do each of the following. Try to avoid using loops.

   a. Plot all the intensities in A, sorted in decreasing value. (Note, in this case we don't care about the 2D structure of A, we only want to sort all the intensities in one list.)
   b. Create and display a new color image the same size as A, but with 3 channels to represent R G and B values. Set the values to be bright red (i.e., R = 255) wherever the intensity in A is greater than a threshold t, and black everywhere else.
   c. Create a new image X that consists of the top left quadrant of A.
   d. Generate a new image, which is the same as A, but with A's mean intensity value subtracted from each pixel.
   e. Let y be the vector: y = [1: 12]. Use the reshape command to form a new matrix z whose first column is [1, 2, 3, 4]', whose second column is [5, 6, 7, 8]' and whose third column is [9, 10, 11, 12]'.
   f. Let v be the vector: v = [1 8 8 2 1 3 9 8]. Set a new variable x to be the number of 8's in the vector v.

II. Short programming example [60 points]

Write functions to do each of the following to an input grayscale image, and then write a script that loads an image, applies each transformation to the original image, and displays the results in a figure using the Matlab subplot function. Label each subplot with title.

(Each 10 points) Apply the script to an image(s) of your choosing, and show us the results.

a. Flip the image vertically.
b. Flip the image horizontally.
c. map a grayscale image to its "negative image", in which the lightest values appear dark and vice versa.
d. swap the red and green color channels of the input color image.
e. average the input image with its mirror image (use typecasting!)
f. add or subtract a random value between [0,255] to every pixel in a grayscale image, then clip the resulting image to have a minimum value of 0 and a maximum value of 255.

Matlab tips: Do the necessary typecasting (uint8 and double) when working with or displaying the images. Some useful functions: title, subplot, imshow, mean, imread, rgb2gray (converts a color image to grayscale). Do NOT use built-in functions of the related libraries (except loading and displaying images). Write your own algorithms.

- Finally, archive the following files and submit via ESTUOYS before the deadline.
  - MS Word file including your answer from the first part of this homework.
  - *.m file you wrote.
- Each group (consisting of 2 students) should make single submission. You can not change your groups during the semester.
- Indicate IDs and names of group members in the submission.